Jianhong Tu

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Education

University of California, Santa Cruz, PhD in Computer Science — Sept 2025 – Present

- **Advisor:** Chenguang Wang

Washington University in St. Louis, PhD in Computer Science — Jan 2025 – Jul 2025

- **Advisor:** Chenguang Wang

Washington University in St. Louis, BS in Data Science — Sept 2021 – Dec 2024

- GPA: 4.0/4.0

Publications

[1] **Jianhong Tu**, Zhuohao Ni, Nicholas Crispino, Zihao Yu, Michael Bendersky, Beliz Gunel, Ruoxi Jia, Xin Liu, Lingjuan Lyu, Dawn Song, and Chenguang Wang. 2025. MLAN: Language-Based Instruction Tuning Preserves and Transfers Knowledge in Multimodal Language Models. In *Proceedings of the 3rd Workshop on Towards Knowledgeable Foundation Models (KnowFM at ACL 2025)*.

[2] Kyle Montgomery, David Park, **Jianhong Tu**, Michael Bendersky, Beliz Gunel, Dawn Song, and Chenguang Wang. 2025. Predicting Task Performance with Context-aware Scaling Laws. In *Proceedings of the 3rd Workshop on Towards Knowledgeable Foundation Models (KnowFM at ACL 2025)*.

[3] Ye Yuan, Junyu Luo, Guancheng Wan, Jinsheng Huang, Chengwu Liu, Junwei Yang, Yifang Qin, Zhiping Xiao, Qingqing Long, Meng Xiao, Yiqiao Jin, **Jianhong Tu**, Yuqi Chen, Wei Ju, Zhongwei Wan, Yusheng Zhao, Xiao Luo, Yiwei Fu, Yizhou Sun, Wei Wang, and others. 2025. A Comprehensive Survey of Evaluating Multimodal Foundation Models: Hierarchical Perspective and Extensive Applications. *Preprint*, May 2025.

[4] **Jianhong Tu** and Anxu Wang. 2024. Diversity-based Data Subset Selection with Deep Reinforcement Learning. *Preprint*, December 2024.

[5] **Jianhong Tu**, Erdong Chen, and Shuhan Zhang. 2024. Instruction-aware Visual Feature Extraction for Multimodal Large Language Model. *Preprint*, December 2024.

[6] James N. McGregor, Clayton A. Farris, Sahara Ensley, Aidan Schneider, Leandro J. Fosque, Chao Wang, Elizabeth I. Tilden, Yuqi Liu, **Jianhong Tu**, Halla Elmore, Keenan D. Ronayne, Ralf Wessel, Eva L. Dyer, Kiran Bhaskaran-Nair, David M. Holtzman, and Keith B. Hengen. 2024. **Failure in a population: Tauopathy disrupts homeostatic set-points in emergent dynamics despite stability in the constituent neurons.** *Neuron*, Aug 2024. (COVER PAPER)

Experience

Research Assistant, Hengen Lab – Washington University in St. Louis — May 2022 – Dec 2024

Advisors: Keith Hengen and Aidan Schneider

- Built a transformer-based seq2seq action recognition model improving accuracy over ensemble XGBoost.
- Built and trained neural decoding models that infer light–dark cycles from large-scale mouse neural recordings to identify environment-responsive neural patterns.

Research Fellow, BioSURF – Washington University in St. Louis — May 2023 – Sept 2023

Advisor: James N. McGregor

- Proposed a novel CNN-Transformer model achieving over 95% accuracy in mouse sleep-wake analysis.
- Conducted statistical analysis on the relationship between sleep deficiency and tauopathy.

Teaching Experience

Peer Research Mentor, Research Experience for Undergraduates Program – WashU	June 2024 – Dec 2024
Teaching Assistant, CSE 527: Natural Language Processing – WashU	Jan 2024 – May 2024
Teaching Assistant, CSE 561: Large Language Models – WashU	Jan 2024 – May 2024
Teaching Assistant, CSE 131: Introduction to Computer Science – WashU	Sept 2021 – May 2022

Awards and Honors

Dean's Select PhD Fellowship – WashU	2025
Summa Cum Laude – WashU	2024
Senior Excellence Award – WashU	2024
Antoinette Frances Dames Award – WashU	2022 – 2024
Biology Summer Undergraduate Research Fellowship – WashU	2022
Dean's List – WashU	2021 – 2024